

Recall from previously:

- The subset $\{v_1, v_2, \dots, v_p\}$ of the vector space V is **linearly independent** if

$$c_1v_1 + \dots + c_pv_p = 0_V$$

has only the trivial solution.

- **Theorem:** Given $\{v_1, \dots, v_p\}$ with $p \geq 2$ and $v_1 \neq 0_V$. The vectors $v_1, \dots, v_p$ are linearly dependent if and only if for some $j > 1$, the vector v_j is a linear combination of $v_1, \dots, v_{j-1}$.
- Let H be a subspace of a vector space V . The set $\{v_1, \dots, v_p\}$ is a **basis** for H if
 1. $\{v_1, \dots, v_p\}$ is linearly independent,
 2. $H = \text{Span}\{v_1, \dots, v_p\}$.
- **Spanning Set Theorem:** Let $S = \{v_1, \dots, v_p\} \subseteq V$ and $H = \text{Span}\{v_1, \dots, v_p\}$.
 1. Suppose v_k can be written as a linear combination of the other vectors in S , then $S - \{v_k\}$ still spans H .
 2. If $H \neq \{0_V\}$, some subset of S is a basis for H .
- **Theorem:** The pivot columns of an $m \times n$ matrix A form a basis for $\text{Col}(A)$.

Theorem: Let $\mathcal{B} = \{b_1, \dots, b_n\}$ be a basis for a vector space V . Then for each $x \in V$ there exists unique scalars $c_1, \dots, c_n$ such that $x = c_1b_1 + \dots + c_nb_n$.

1. Prove the theorem.

Definition 1. Suppose $\mathcal{B} = \{b_1, \dots, b_n\}$ is a basis for a vector space V . The **coordinates of x relative to $\mathcal{B}$** are the weights $c_1, \dots, c_n$ such that $x = c_1b_1 + \dots + c_nb_n$. We define the **coordinate mapping** from V to $\mathbb{R}^n$ by $[x]_{\mathcal{B}} = \begin{bmatrix} c_1 \\ \vdots \\ c_n \end{bmatrix}$.

2. Let $V = \mathbb{R}^3$ with basis $\mathcal{E} = \{e_1, e_2, e_3\}$. Let $x = 3e_1 + 4e_2 + 5e_3$, then find $[x]_{\mathcal{E}}$. Let $\mathcal{B} = \left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \right\}$. Find $[x]_{\mathcal{B}}$.

3. Let $b_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ and $b_2 = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$, then $\mathcal{B} = \{b_1, b_2\}$ is a basis for $\mathbb{R}^2$. Let $x = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$. Find $[x]_{\mathcal{B}}$.

Definition 2. Let $\mathcal{B} = \{b_1, \dots, b_n\}$, then define $P_{\mathcal{B}} = [b_1 \ \dots \ b_n]$. We call $P_{\mathcal{B}}$ the **change of coordinates matrix**.

Note $x = P_{\mathcal{B}}[x]_{\mathcal{B}}$ and $[x]_{\mathcal{B}} = P_{\mathcal{B}}^{-1}x$.

Theorem: Let $\mathcal{B} = \{b_1, \dots, b_n\}$ be a basis for a vector space V . Then the coordinate mapping $x \mapsto [x]_{\mathcal{B}}$ is a one-to-one, onto, linear transformation from V to $\mathbb{R}^n$.

4. Prove the theorem.

Definition 3. Let $T : V \rightarrow W$ be a linear map of vector spaces. If T is one-to-one and onto, then T is a **vector space isomorphism**, and we say V and W are isomorphic. We write $V \cong W$.

5. Show that $\mathbb{P}_3$ and $\mathbb{R}^4$ are isomorphic.

Theorem: If a vector space V has a basis $\mathcal{B} = \{b_1, \dots, b_n\}$, then any set in V containing more than n vectors is linearly dependent.

6. Prove the theorem.

Cor: If V is a vector space with basis $\mathcal{B} = \{b_1, \dots, b_n\}$, then each set of linearly independent vectors has at most n elements.

Theorem: If a vector space V has a basis with n elements, then every basis of V must have exactly n vectors.

7. Prove the theorem.

Definition 4. If V is spanned by a finite set, then V is called **finite dimensional**, and the **dimension** of V , written $\dim(V)$, is the number of vectors in a basis of V . Define $\dim(\{0_V\}) = 0$. If V is not spanned by any finite set, then V is called **infinite dimensional**.